

1) Selection Sort -

Aims - To apply the Selection Sort algorithm on the Dataset [64, 25, 12, 22, 11] using the brute force technique

Problem Interpretation - Selection Sort is a Comparison based sorting algorithm that repeatedly selects the smallest element

Algorithm

1. Assume the first element is the minimum
2. Compare it with the remaining elements
3. Find the smallest element
4. Swap it with the first unsorted element
5. Repeat until the array is sorted

Dataset - [64, 25, 12, 22, 11]

Sorting Process

Pass
~~Initial~~

Array

64, 25, 12, 22, 11

Pass 1

11, 25, 12, 22, 64

Pass 2

11, 12, 25, 22, 64

Pass 3

11, 12, 22, 25, 64

Pass 4

11, 12, 22, 25, 64

Number of Comparisons

$4+3+2+1=10$ Comparisons

Time Complexity

- Best case: $O(n^2)$
- Average case: $O(n^2)$
- Worst case: $O(n^2)$

Final Output - $[11, 12, 22, 25, 64]$

Result - The Selection Sort Algorithm Successfully Sorted the given Dataset in ascending order with 10 Comparisons

2) Closest Pair - Brute force Analysis

Aim - To Determine the Closest pair of points using the Brute force closest pair algorithm

Problem Interpretation - The closest pair problem aims to identify the two points with the minimum euclidean distance

Given points $(2, 3), (12, 8), (10, 5), (5, 1), (12, 10), (3, 4)$

Method

The Distance between every pair is Calculated using:

$$\text{Distance} = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

is obtained between:

$(2,3)$ and $(3,4)$

$$\text{Distance} = \sqrt{(2-3)^2 + (3-4)^2}$$
$$= \sqrt{2} \approx 1.414$$

Number of Comparisons

for n points

$$n(n-1)/2$$

$$= \frac{6 \times 5}{2}$$

$= 15$ Comparisons

Time Complexity

- Best Case: $O(n^2)$
- Average Case: $O(n^2)$
- Worst Case: $O(n^2)$

Result :- The closest pair is $(2,3)$ and $(3,4)$ with a minimum Distance of 1.414 units

3) Convex Hull - Brute force Approach

Aim :- To Determine the Convex Hull of the given Set of points using the Brute force Algorithm

Problem interpretation

A Convex Hull is the smallest convex polygon that encloses all the given points. The Brute force method checks every pair of points and determine

Given Points $(0,3), (2,2), (1,1), (2,1), (3,2), (0,2), (3,3)$

Procedure

1. Draw all possible line segments
2. Check whether all remaining points lie on one side
3. if true, the line belongs to the Convex hull
4. Repeat until all hull edges are identified

Convex Hull points - $(0,2), (3,2), (3,3), (0,3)$

Comparison

Number of point pairs
 $= 7 \times 6 / 2$

$= 21$ pairs checked

Time Complexity - $O(n^2)$

Result - The Outer Boundary consists of the points $(0,2), (3,2), (3,3)$ and $(0,3)$

4) Searching Techniques - Comparative Analysis

Aim - To Compare Linear Search and Binary Search

Problem interpretation - Searching algorithms are used to locate an element in a dataset. Linear Search scans each element one by one, while Binary Search repeatedly divides a sorted array into halves.

Linear Search

- (3, 3)
- Works on both sorted & unsorted data
 - Searches sequentially
 - Best Case = $O(1)$
 - Average Case = $O(n)$
 - Worst Case = $O(n)$

Binary Search

- Works only on sorted data
- Divides the search space into two halves
- Best Case = $O(1)$
- Average Case = $O(\log n)$
- Worst Case = $O(\log n)$

Result - Binary Search is more efficient for sorted dataset, whereas linear search is preferred for small or unsorted dataset

5) Closest Pair - Real time Scenario

Aim :- To identify the nearest emergency centers using the brute force closest pair algorithm.

Problem Interpretation - Emergency response system require identifying the nearest service centers to reduce response time. The brute force method compares every possible pair of GPS coordinates and determines the minimum distance.

Procedure

1. Consider all emergency center coordinates
2. Compute the euclidean distance between every
3. Compare all distances
4. Select the pair with the smallest distance

Number of Comparisons

For n centers:

$n(n-1)/2$ pairwise comparisons

Time Complexity

- Best Case: $O(n^2)$
- Average case: $O(n^2)$
- Worst Case: $O(n^2)$

Applications

- Ambulance routing
- Fire station planning
- Disaster management
- GPS navigation
- Emergency Response Systems

Result

Brute force closest pair algorithm successfully

identifies the nearest emergency centers by

comparing all possible pairs. Although it guarantees

the correct result, its $O(n^2)$ time complexity

makes it less suitable for very large datasets